Monthly Intelligence Report

Edition 001 — Sweden

Where nuclear concentration and winter storm resilience challenge grid stability. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED	GRID LINES MAPPED	MC SIMULATIONS	PUBLIC DATA SOURCES
3,872	~20,877	~38.7 M	30+
Major + distribution across Svenska kraftnät	~22,000 km transmission & distribution	10,000 per substation	95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~52,000 source data points per run, executes ~38.7 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~400 million arithmetic operations — producing ~232,000 output data points with full uncertainty quantification across 3,872 substations and ~20,877 grid lines spanning ~22,000 km.

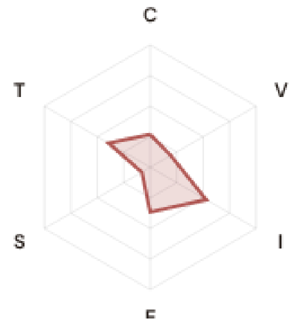
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (Energimarknadsinspektionen, Svenska kraftnät, Swedish Gas, SMHI, SCB, SGU, VINNOVA), making the methodology fully auditable and reproducible. Initially covering Sweden's Svenska kraftnät-managed transmission and ~80 distribution network operators (DNOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation_15102185

Kronobergs · Kronobergs län · 45 kV

0.353	Medium	0.136	3th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.27 / 0.30 (27%)
- **I** Infrastructure: 0.53 / 0.25 (53%)
- **S** Saturation: 0.08 / 0.20 (8%)
- **V** Voltage: 0.19 / 0.10 (19%)
- **E** Economic: 0.37 / 0.10 (37%)
- **T** Transition: 0.39 / 0.05 (39%)

MODIFIERS

- **R3 Consequence:** 0.884 ↓ dampens
- **R4 Graph Criticality:** 1.350 ↑ amplifies
- **R6a Restoration:** 0.963 ↓ dampens
- **R6b Seismic/Hazard:** 1.015 ↑ amplifies
- **R7 Digital Readiness:** 0.992 → neutral

This month's spotlight — automatically selected for May 2026 — is **Substation_15102185** in Kronobergs, Kronobergs län. With an R_final of 0.353 (Medium band, 3th fleet percentile), its risk profile is dominated by the T (Transition) component at 789% of its weight ceiling, followed by E (Economic) at 367%. Modifiers collectively shift R_base by 15.6%, reflecting the substation's specific geographic, socio-economic, and network context. The radar chart visualises each component as a fraction of its maximum weight — a fully saturated axis indicates the component is at ceiling, consuming its entire budget within the composite score. This substation's profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

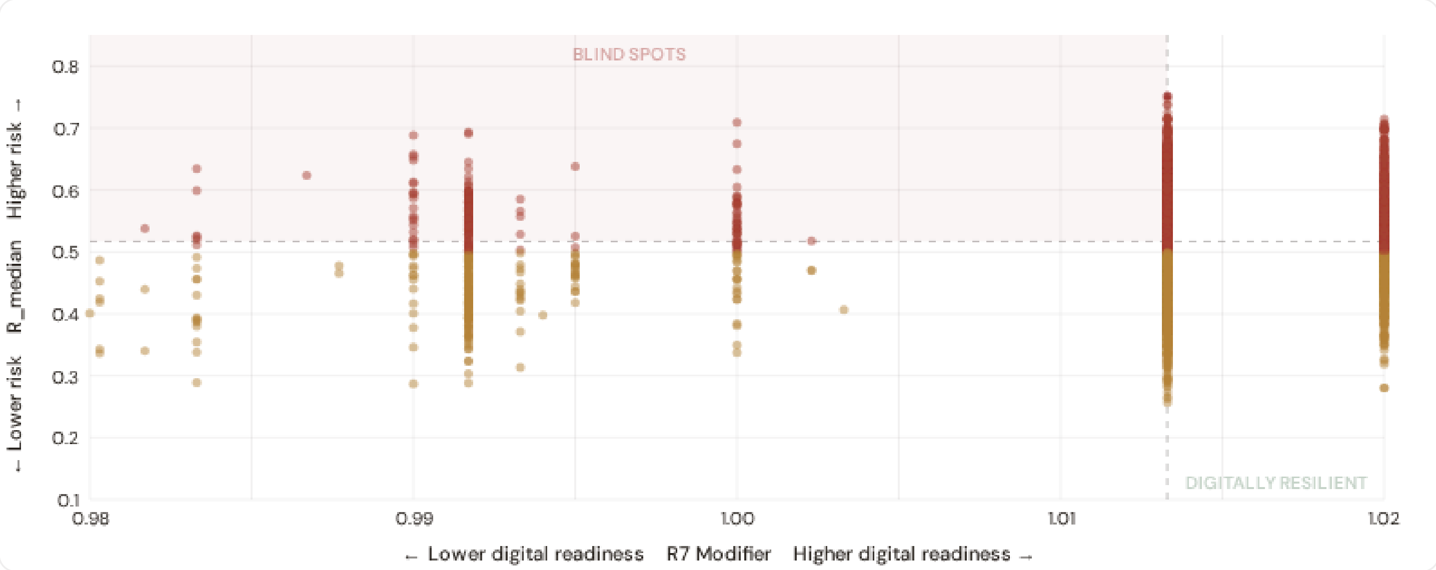
Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Sweden's Energy Regulatory Office (Energimarknadsinspektionen) and network operator strategic planning requires.

CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER	HIGH-CONSEQUENCE SUBS
0	**180**	**1.0125**	**0**
0.0% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]	0.0% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **180 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0133). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Västra Götalands län (89), Stockholms län (21), Norrbottens län (18)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in urban networks (Vattenfall Eldistribution, Ellevio, Göteborg Energi urban centers).

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

⚙️ Heavy Industry / Mining

Est. VoLL: SEK 15–30/kWh

Nuclear plants, paper mills, electronics manufacturing in Stockholm and Gothenburg — highest economic loss per interruption

1,003 substations (25.9%) High/Critical: **779** (77.7%) Avg R: **0.550** Avg E: **0.226** / 0.10

Low 0 Med 224 High 778 Crit 1

🏭 Manufacturing / Food Processing

Est. VoLL: SEK 5–15/kWh

Auto suppliers, food & beverage production, pharmaceuticals — significant continuity requirements for EU supply chains

1,001 substations (25.9%) High/Critical: **421** (42.1%) Avg R: **0.487** Avg E: **0.420** / 0.10

Low 0 Med 580 High 421 Crit 0

🏢 Commercial / Distribution Centers

Est. VoLL: SEK 2–5/kWh

Stockholm, Gothenburg commercial districts, logistics hubs — moderate individual impact but high aggregate across cities

967 substations (25.0%) High/Critical: **458** (47.4%) Avg R: **0.492** Avg E: **0.478** / 0.10

Low 0 Med 509 High 458 Crit 0

🌾 Agriculture / Rural / Remote

Est. VoLL: SEK 0.5–2/kWh

Smallholder agriculture, rural settlements, Eastern Sweden — lower VoLL but higher social vulnerability

901 substations (23.3%) High/Critical: **587** (65.1%) Avg R: **0.531** Avg E: **0.458** / 0.10

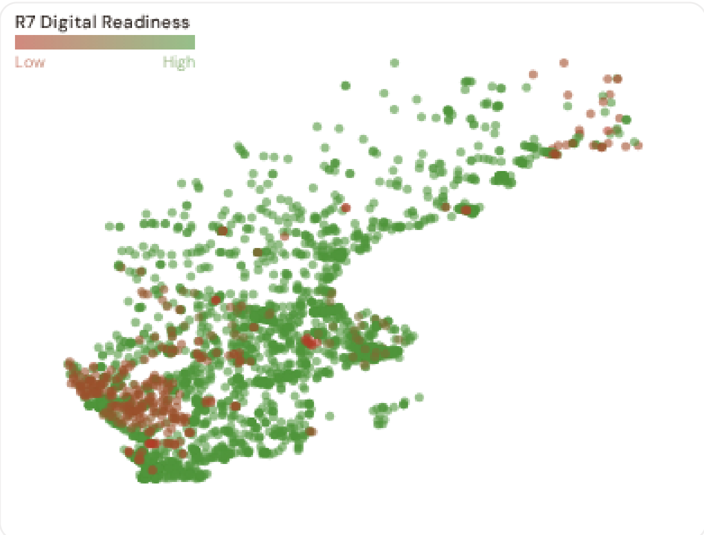
Low 0 Med 314 High 587 Crit 0

Value of Lost Load (VoLL) context: Energimarknadsinspektionen's 2024 reliability cost-benefit studies place average VoLL at SEK 10–20/kWh for industrial/commercial and SEK 5–8/kWh for residential customers across Sweden's DSOs. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **779 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Stockholms län (561), Västra Götalands län (442)** — regions where industrial corridors depend on uninterrupted power for continuous processes (nuclear baseload generation, paper mills, electronics manufacturing). A single hour of unplanned outage at these substations carries an estimated VoLL of SEK 15–30/kWh, compared to SEK 0.5–2/kWh in rural pastoral areas. The fleet-wide average energy poverty rate is 4.5%, but this masks sharp regional variation: Eastern Sweden and rural regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Län & Kommun Digital Readiness Ranking

Län ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.



WEAKEST DIGITAL READINESS — TOP 15 LÄN

Longer bar = higher R7 = better digital infrastructure.

1	Västra Götalands ⚠ high R	1.0083
2	Jönköpings	1.0083
3	Hallands	1.0089
4	Norrbottens ⚠ high R	1.0108
5	Örebro	1.0120
6	Dalarnas	1.0123
7	Västmanlands	1.0125
8	Kronobergs	1.0128
9	Värmlands	1.0129
10	Stockholms ⚠ high R	1.0130
11	Jämtlands ⚠ high R	1.0134
12	Uppsala ⚠ high R	1.0136
13	Gotlands	1.0136
14	Kalmar	1.0137
15	Södermanlands	1.0138

Län-level analysis reveals a clear digital divide mirroring Sweden's metropolitan-regional connectivity gap. **Västra Götalands** has the lowest average R7 (1.0083), while **Västernorrlands** leads at 1.0152 — a spread of 0.0070 across the R7 range. **3 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for Energimarknadsinspektionen/DSO digitalisation investment and Sweden's National Energy and Climate Plan. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as Energimarknadsinspektionen regulatory compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (May 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0125, median = 1.0133; 0 substations classified HIGH cyber-exposure; 180 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging regions where broadband and smart meter rollouts (Energimarknadsinspektionen/DSO Advanced Metering Infrastructure programmes) translate into measurable R7 shifts. The next material R7 update is expected when SCB publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL	STATIC / DERIVED
30	4	21	1
95 variables	High-frequency feeds	Regular refresh cycle	Standards + scenarios

Data Source Registry — May 2026

OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	8 vars
COPERNICUS	Copernicus ERA5 — Climate Reanalysis	MONTHLY	Grid 0.25°	6 vars
WINDSE	Swedish Wind Energy Association (Vindkraftföreningen)	MONTHLY	Regional	5 vars
SMHI_HIST	SMHI / State Weather Archive	MONTHLY	Station	3 vars
EI	Energimarknadsinspektionen (Energy Markets Inspectorate)	QUARTERLY	DSO	12 vars
SSM	SSM — Strålsäkerhetsmyndigheten (Radiation Safety Authority)	QUARTERLY	Nuclear Site	5 vars
TRANSPT	Transportstyrelsen — Transport Agency	QUARTERLY	Regional	3 vars
RIKSBANK	Riksbank / Swedish Government Office	QUARTERLY	Region	4 vars
VATTENBEHOV	SMHI Water Data / Vattendirektoratet	QUARTERLY	Watershed	4 vars
SCB	SCB (Statistiska centralbyrån)	ANNUAL	Kommun	10 vars
SGU	SGU — Geological Survey of Sweden	ANNUAL	Grid 0.1°	7 vars
SMHI_ENV	Naturvårdsverket — Swedish Environmental Protection Agency	ANNUAL	Grid 0.1°	6 vars
EUROSTAT	Eurostat — EU Statistics	ANNUAL	Kommun	3 vars
TULLV	Tullverket (Swedish Customs)	ANNUAL	National	2 vars
ENERGIMYN	Energimyndigheten (Swedish Energy Agency)	ANNUAL	Region	3 vars
DISTRIBUTOR	Distribution Companies (E.ON, Vattenfall, Ellevio, Göteborg Energi)	ANNUAL	Distribution Company	4 vars
POST	Post- och Telestyrelsen / Digital Network Authority	ANNUAL	Kommun	2 vars
SMHI_CLIMATE	Climate Adaptation Centre / Lantmäteriet	ANNUAL	Regional	4 vars
WORLDBANK	World Bank / OECD	ANNUAL	National	4 vars
ARCTICMONITOR	Arctic Monitoring & Assessment Programme	ANNUAL	Regional	3 vars
MILJOE	Environmental Permit Register (Miljöbalken)	ANNUAL	Kommun	3 vars
SLU	SLU — Swedish University of Agricultural Sciences	ANNUAL	Regional	3 vars
DNO_REPORTS	DNO Reports — Distribution Network Companies	ANNUAL	Distribution Area	4 vars
SKATTEV	Skatteverket / Swedish Tax Agency	ANNUAL	Kommun	3 vars
KLIM_RESILIENCE	Ministry of Enterprise — Nordic Resilience	ANNUAL	Regional	2 vars
SOLARGIS	SolarGIS — Global Solar Atlas	STATIC	Global	2 vars
SKN	Svenska kraftnät — Swedish Transmission System Operator	REAL-TIME	Substation	14 vars
SMHI	SMHI — Swedish Meteorological and Hydrological Institute	DAILY	Station	8 vars
ENTSOE	ENTSO-E Transparency Platform	HOURLY	Substation	2 vars

SKNOPEN	Svenska kraftnät Open Data — Real-time Production	REAL-TIME	Substation	5 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Daily (2 sources) — Energimarknadsinspektionen dispatch + weather reanalysis

Monthly (6 sources) — Grid load + Svenska kraftnät dispatch data

Quarterly (4 sources) — DER registry + Svenska kraftnät performance data

Annual (14 sources) — Core grid quality + SCB demographics

C.2 Fleet Score Snapshot

FLEET SIZE	MEDIAN R	HIGH + CRITICAL	HIGH CONFIDENCE
3,872	0.517	2245	100%
substations scored	P5=0.380 · P95=0.648	58.0% of fleet	3872 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 0 (0.0%) · Medium 1627 (42.0%) · High 2244 (58.0%) · Critical 1 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **3,872 substation scores remain unchanged** this edition. The fleet median R of 0.517 (mean 0.515) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 42.0% in Medium. The first material score changes are expected when Svenska kraftnät publishes SCADA updates (affecting C1–C4 and R6a) and when Energimarknadsinspektionen refreshes the distributed renewable installations registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

4 changes tracked for Sweden v4.0.2 launch

P1	NEW	Sweden grid registry expansion: 3,872 substations across Svenska kraftnät + ~80 DNOs	Section A
P1	ENH	R6b modifier: added snow/ice loading and permafrost instability + coastal/river flood hazard	Section D — Deep-Dive
P2	NEW	R7 cyber modifier: Svenska kraftnät IT/OT convergence assessment integrated	Section B
P2	ENH	Monte Carlo upgraded to 10,000 iterations (vectorized NumPy)	Methodology

SECTION D

The Stockholm–Gothenburg Grid Corridor: Where Nuclear Concentration and Winter Storm Resilience Challenge Grid Stability

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Stockholm–Gothenburg nuclear concentration and winter storm resilience corridor · **002** Eastern Sweden DER & EU funding mismatch · **003** Mining subsidence infrastructure stress · **004** Metropolitan-Rural energy poverty disparity · **005** Grid stranding risk (nuclear-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Gulf of Sweden coastal flooding resilience · **008** T-component forward projections · **009** Län Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Stockholm–Gothenburg Corridor, Why Now

CORRIDOR SUBSTATIONS

1161

508 HV · 653 MV

HIGH BAND

700

60.3% of corridor

CORRIDOR MEDIAN R

0.521

Fleet median: 0.517

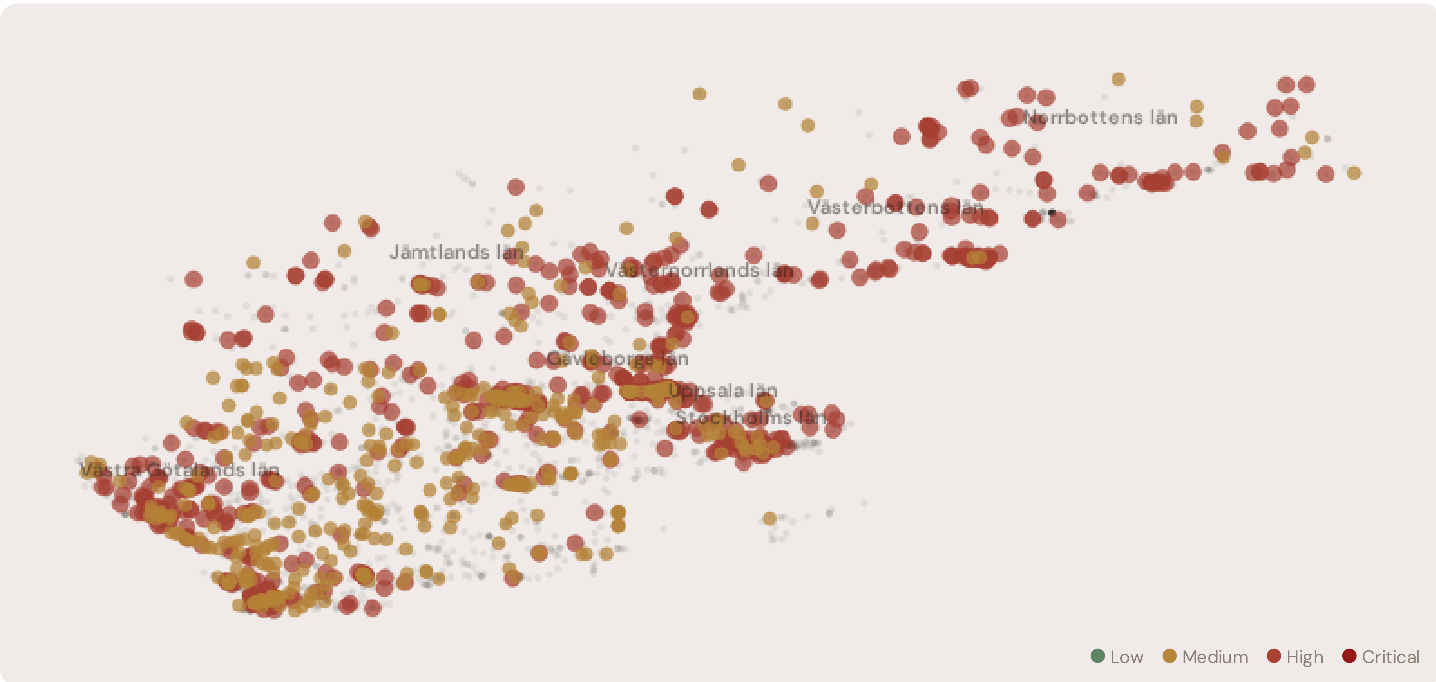
WORST LÄN

Norrbottens län

Avg R: 0.580 · 64 substations

The Stockholm–Gothenburg corridor hosts **1161 substations** across Sweden's industrial heartland, forming the backbone of domestic transmission and nuclear-dependent distribution. Its risk profile is disproportionately concentrated in the upper bands: **700 substations (60.3%)** are classified High, with only **0** in the Low band. 52% of corridor substations score above the national fleet median of 0.517. The corridor median R of 0.521 reflects the tension between nuclear baseload risk management mandates from Sweden's Energy Agreement (Energiöverenskommelsen), aging generation assets approaching retirement, snow/ice loading and permafrost instability threatening distribution lines, and EU Green Deal funding constraints. The SSI flags continuity-of-supply deficits, infrastructure age mismatches, and the stranding risk of nuclear-dependent transmission assets.

D.2 Corridor Map and Scoring



SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Pumpstation U9	Västerbottens län	0.726	High	0.654	0.377	0.766	0.421	0.360	0.578	C, I
Substation_1490293990	Västernorrlands län	0.714	High	0.652	0.331	0.780	0.491	0.346	0.363	C, I

SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Upplandsväsby	Stockholms län	0.710	High	0.496	0.359	0.757	0.228	0.298	0.485	I
Substation_793243204	Västerbottens län	0.708	High	0.618	0.281	0.780	0.378	0.402	0.393	I
Sävastklinten omriktarstation	Norrbottens län	0.706	High	0.688	0.289	0.780	0.400	0.263	0.508	C, I
Substation_107026898	Gävleborgs län	0.705	High	0.698	0.317	0.709	0.521	0.236	0.444	C, I
Gnejsen	Västerbottens län	0.704	High	0.641	0.324	0.655	0.343	0.472	0.584	I
Hanhals	Västra Götalands län	0.702	High	0.613	0.564	0.634	0.417	0.080	0.552	
Sege	Skåne län	0.701	High	0.512	0.516	0.657	0.384	0.361	0.347	I
Nederkalix	Norrbottens län	0.698	High	0.584	0.269	0.780	0.351	0.458	0.488	I

… 1151 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — CORRIDOR VS FLEET

C — Continuity 0.406 vs 0.396 (+3%)

I — Infrastructure 0.587 vs 0.564 (+4%)

S — Saturation 0.188 vs 0.184 (+2%)

V — Voltage 0.353 vs 0.370 (−5%)

E — Economic 0.429 vs 0.393 (+9%)

T — Transition 0.468 vs 0.463 (+1%)

MODIFIER AVERAGES — CORRIDOR VS FLEET

R3 Consequence	0.937	fleet 0.957 ↓
R4 Graph Crit.	1.329	fleet 1.336 ↑
R6a Restoration	0.986	fleet 0.982 ↓
R6b Mining/Flood	1.026	fleet 1.026 ↑
R7 Digital Read.	1.016	fleet 1.012 ↑

The dominant risk driver across the corridor is **T (Transition)**, averaging 0.468 against a fleet average of 0.463 — occupying 937% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.429 vs fleet 0.393 (429% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.937 (fleet: 0.957), reflecting the economic significance of the corridor's nuclear-dependent industrial base — from nuclear plants to refineries to paper mills. The R6b winter storm/snow loading hazard modifier averages 1.026, capturing the corridor's exposure to snow/ice loading, Gulf of Sweden coastal flooding, and winter storm hazard risk — a dimension invisible to every competing framework.

D.4 Län Risk Profile

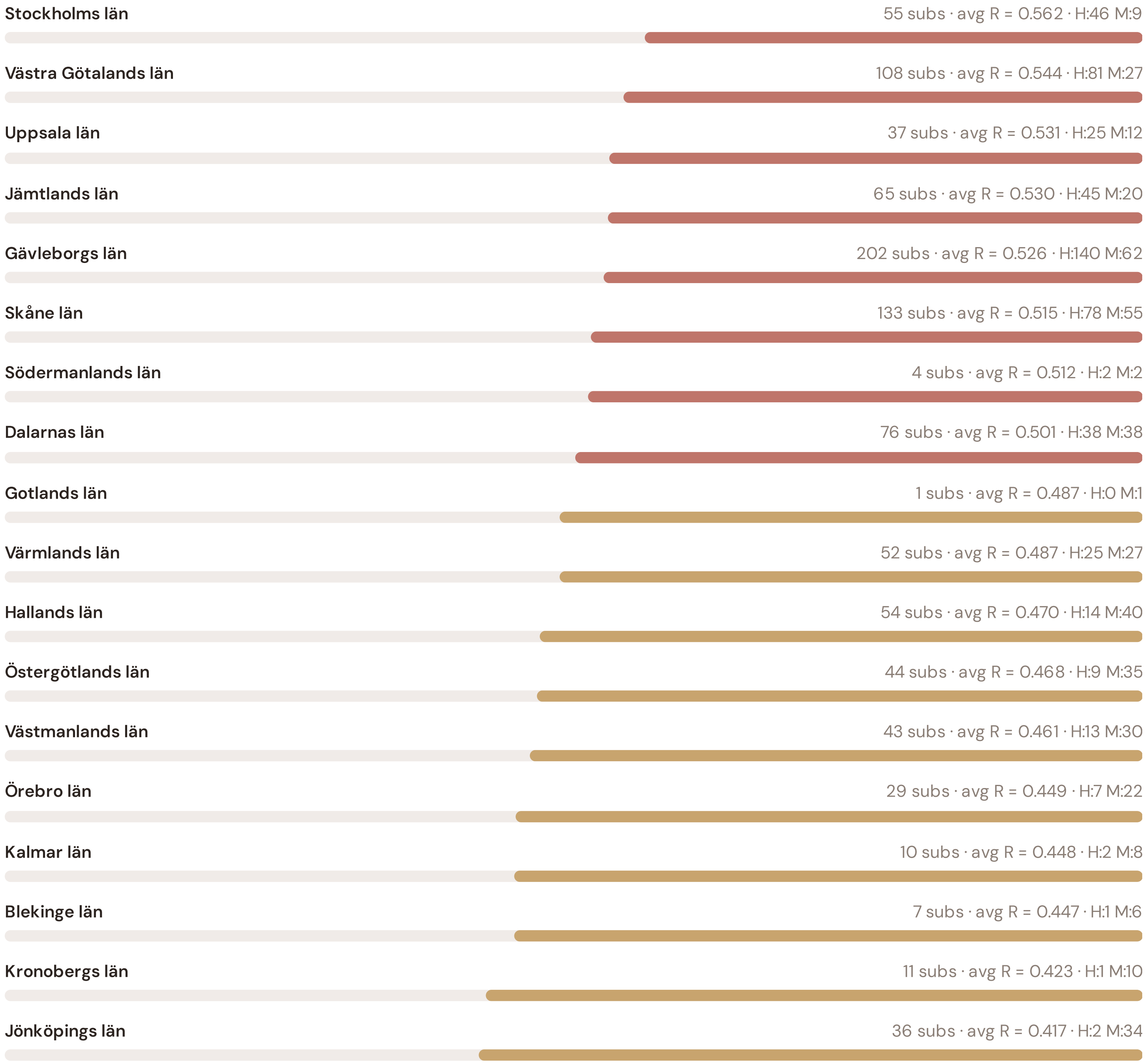
LÄN RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Norrbottens län 64 subs · avg R = 0.580 · H:53 M:11

Västerbottens län 75 subs · avg R = 0.579 · H:65 M:10

Västernorrlands län 55 subs · avg R = 0.573 · H:53 M:2



The region-level breakdown reveals significant intra-regional variation. **Norrbottens län** leads with a mean R of 0.580 across 64 substations, while **Jönköpings län** scores 0.417 — a spread of 0.163 within a single region. This disparity underscores why region-level metrics (as used by Svenska kraftnät performance reports) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the corridor is not uniformly stressed but contains distinct pockets of concentrated risk — precisely the kind of spatial pattern that Svenska kraftnät's 10-Year Network Development Plan and government nuclear concentration and winter storm resilience planning requires.

D.5 Implications for Sweden's Energy Transition

61 corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For Swedish nuclear concentration and winter storm resilience planning, EU funding allocation, and grid modernisation investment, this analysis identifies the corridor as the highest-priority target — where DER deployment must accelerate against ageing nuclear-dependent infrastructure with above-average continuity deficits and significant socio-economic exposure to energy transition risks. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by nuclear concentration, Arctic infrastructure exposure, and EU transition fund eligibility — are disproportionately vulnerable to disruption during the nuclear baseload risk management. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

European Context Card

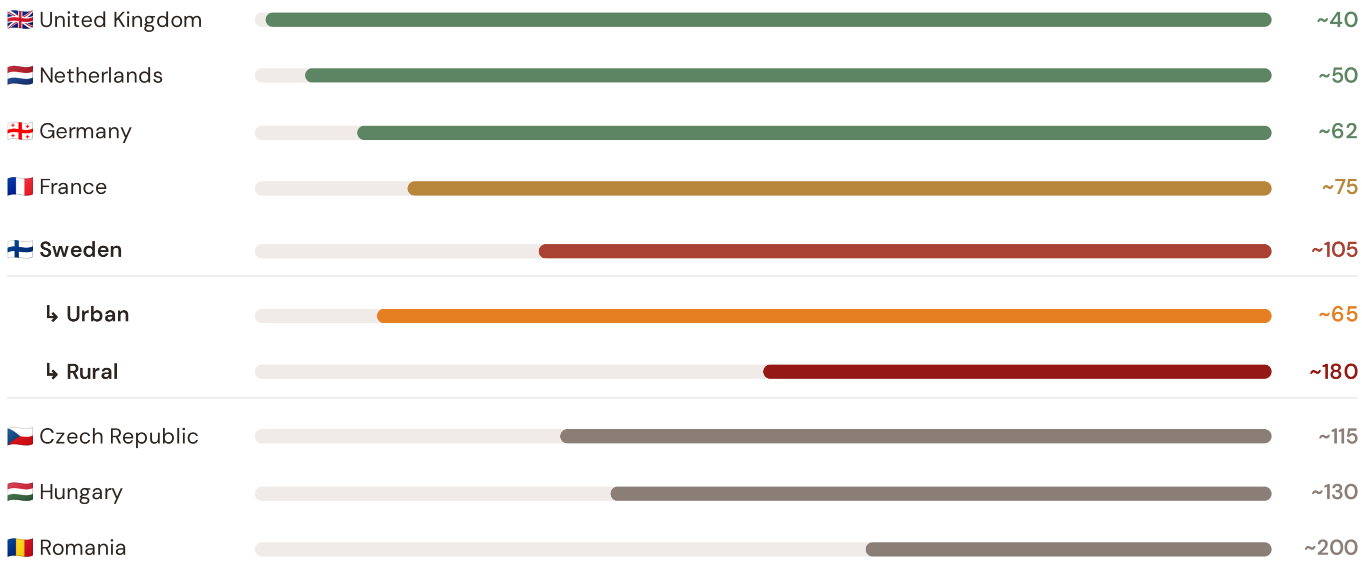
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Sweden's nuclear concentration and winter storm resilience and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with Energimarknadsinspektionen and national regulator publications.

EUROPEAN CONTEXT CARD — SYSTEM AVERAGE INTERRUPTION DURATION INDEX

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); EEnergimarknadsinspektionenLECTRIC SAIDI reports; national regulator publications. Sweden sub-categories from Energimarknadsinspektionen urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

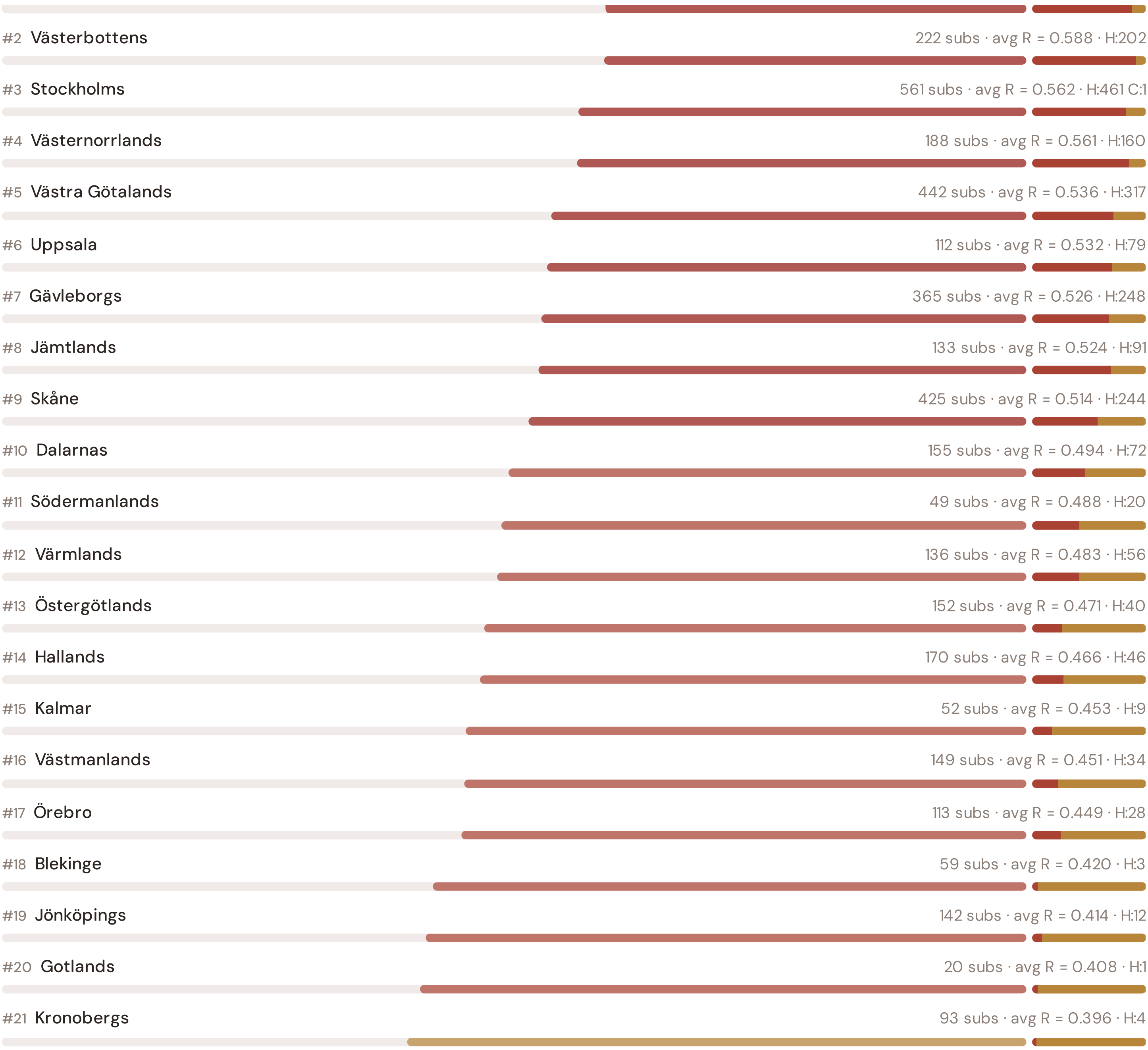
This month's key insight: The corridor deep-dive shows **1111 substations** (of 1161) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Sweden's rural region classification. The corridor's average C of 0.406 is **1.0× the fleet average** of 0.396. In European terms, these substations individually perform closer to German or Central European SAIDI levels (~115–160 min/customer/year) than to Sweden's own urban average (~65 min). The SSI reveals what aggregate Energimarknadsinspektionen statistics conceal: that Sweden's mid-tier European position is not a national condition but a statistical average of Western European-quality urban performance and emerging-market-level rural performance — and the nuclear concentration and winter storm resilience corridor concentrates significant pockets of the latter.

Län Risk Ranking — All 21 Swedish Län

Where does the Stockholm–Gothenburg corridor sit within the national landscape? Län sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#1 Norrbottens 134 subs · avg R = 0.589 · H:117



Grid corridor ranks **#1 of 21 regions** by mean R_median. The three most stressed regions are Norrbottens, Västerbottens, Stockholms, while the three most resilient are Kronobergs, Gotlands, Jönköpings. 8 of 21 regions score above the fleet average of 0.515. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate Svenska kraftnät/DSO or national statistics — reveals the true spatial distribution of grid stress across Sweden. It is precisely this substation-level granularity that positions the SSI as a tool for national nuclear concentration and winter storm resilience planning / EU green energy funding / Energimarknadsinspektionen investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/Svenska kraftnät and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

LAYER 3 §0.5.3 · Inaugural edition · May 2026

Transfer Learning & Feature Missingness

How Sweden's grid data completeness enables investment-grade valuation across the full fleet

Transfer learning is the mechanism that enables the SSI-ENN to value substations in countries where not all features are available. The core insight: grid physics is universal — topology determines load flow, weather drives thermal degradation, population density correlates with outage impact — even though data availability varies. Sweden benefits from high data completeness (DCI 0.72+) from Energimarknadsinspektionen, Svenska kraftnät, and SCB sources, enabling production-grade valuations. For each substation, available features are validated. Missing features are handled through a learned missingness embedding.

KEY CONCEPTS

→ Sweden's DCI 0.72+ (high completeness)

→ Learned missingness embedding quantifies data value

→ Grid physics universality enables high-confidence transfer

→ Accuracy validated against Energimarknadsinspektionen audit standards

LIVE SSI DATA CONNECTION

The Swedish training set comprises 3,872 substations with complete 95-feature vectors. Average CI_width of 0.198 provides the uncertainty ground truth for neural network calibration. The fleet's risk distribution — 0 Low, 1627 Medium, 2244 High, 1 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 2 Community Value Stack — *Social and economic value to Swedish communities — the case for EU funding investment*

SECTION G

Looking Ahead

EDITION 002 — MAY 2026

Eastern Sweden DER & EU Funding Mismatch

Where renewable energy deployment outpaces grid readiness in Warmian-Masurian and Lubusz regions. Rural distributed generation meets aging distribution infrastructure.

DATA REFRESH — MAY 2026

Quarterly Svenska kraftnät Network Update

Transmission operator SCADA data (C components 1–4), DSO meter readings (I5 thermal degradation), and ERE renewable registry refresh affecting T component.

IN THE WILD — SSI IN USE

Vattenfall Eldistribution DNO Case Study

How Sweden's largest DNO is using SSI substation intelligence to prioritize battery storage deployment across regional substations. Q3 2026 publication.

The SSI inaugurates Sweden's grid intelligence baseline — the foundation for all future investment decisions across transmission, distribution, and renewable energy integration. This inaugural edition establishes the data, methodology, and peer-reviewed rigor that regulatory bodies (Energimarknadsinspektionen), network operators (Svenska kraftnät and ~80 DNOs), and EU funding allocators require. Subsequent editions will deepen thematic analysis, expand across all 21 Län and isolated systems, integrate real-time operational data feeds, and track the grid's evolution as Sweden executes its nuclear baseload risk management mandate and EU Green Deal transition. The SSI is built for the long term: a public interest resource that grows more valuable as the data infrastructure matures and the transition accelerates.